

4. CV of the researcher

Essaid Chham, PhD

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Current position

Position	Postdoctoral Researcher (APOSTD 2024 programme)		
At UMH since	06/04/2023		
Institution	Miguel Hernández University		
Department / group	Applied Physics	<u>SCOLab</u>	
Country	Spain	Telephone	(+34) 624735931
Keywords	Atmospheric radionuclides; aerosol transport and deposition; DEHM; environmental radioactivity; Monte Carlo modelling.		

Professional experience

Period	Position and institution
10/2024–present	Postdoctoral Researcher, APOSTD 2024 programme, Miguel Hernández University, Spain
04/2023–10/2024	Postdoctoral Researcher, María Zambrano programme, Miguel Hernández University, Spain
2020–2022	Postdoctoral Researcher, Abdelmalek Essaâdi University, Tétouan, Morocco

Education

Degree	University / country	Year
PhD in Physics	Abdelmalek Essaâdi University, Tetuán, Morocco	2019
PhD in Chemistry	Universidad de Granada / Spain	2018
Master in Physics	Abdelmalek Essaâdi University, Tetuán, Morocco	2011
Bachelor in Physics	Abdelmalek Essaâdi University, Tetuán, Morocco	2008

CV summary

I hold PhDs in Physics and Chemistry, with expertise in nuclear science, environmental radioactivity and computational modelling. My early work used Monte Carlo simulations to assess radiation exposure and optimise Morocco's TRIGA Mark II research reactor for medical radioisotope production.

Through an Erasmus Mundus fellowship at the University of Granada (2014–2017), I expanded into atmospheric radionuclides, particularly beryllium-7 (^7Be), combining radiometric measurements, meteorological analysis and machine learning. This work underpins my subsequent research on aerosol transport and atmospheric source–receptor relationships.

Since April 2023, I have worked at Miguel Hernández University (Spain), first under the María Zambrano programme and currently through an APOSTD postdoctoral fellowship (Generalitat Valenciana). My research combines source–receptor analysis with computational modelling of atmospheric radionuclides, including ^7Be and ^{210}Pb .

In collaboration with Zhuyun Ye at Aarhus University, I have contributed to the integration of ^7Be into the Danish Eulerian Hemispheric Model (DEHM), linking cosmogenic production with atmospheric transport, radioactive decay and deposition, and to its evaluation against observations. This development provides the technical basis for STRAT-TRACE's planned source-tagged, controlled experiments. Results from this collaboration were presented at the EGU General Assembly 2026.

Technical skills: DEHM, HYSPLIT, Fortran, R (data.table, fst), Python, NetCDF, Git, Monte Carlo modelling and machine learning.

International mobility: research experience in Morocco, Spain and Italy; three one-month research visits to Aarhus University's Department of Environmental Science in Roskilde, Denmark (March 2025, October 2025 and May 2026), including collaboration with Zhuyun Ye on ^7Be modelling in DEHM.

Research activities

Selected peer-reviewed publications

- E. Chham, J.A.G. Orza. 2025. Evaluating the Impact of the Aerosol Sampling Time Interval on CWT and PSCF Source-Receptor Models: A Critical Perspective. *Sci. Total Environ.* 971: 179069
- A. Nouayti, I. Berriban, E. Chham, et al, 2024. Utilizing innovative input data and ANN modeling to predict atmospheric gross beta radioactivity in Spain. *Atmos. Pollut. Res.* 15(11): 102264.
- A. Nouayti, E. Chham, I. Berriban, et al, 2024. Open-AMA: Open-source software for air masses statistical analysis. *Comput. Geosci.* 189: 105627.
- M. Drissi El-Bouzaidi, T. El Bardouni, E. Chham, et al, 2024. Unfolding of the neutron spectrum in a nuclear reactor using genetic algorithms method. *Nucl. Eng. Des.* 422: 113156.
- I. Berriban, E. Chham, M. Azahra, et al, 2023. Influence of atmospheric circulation and local parameters on activity concentration of gross alpha and gross beta in Granada, Spain. *Atmos. Pollut. Res* 14(9): 101857.
- I. Berriban., M. Azahra, E. Chham, M.A. Ferro-García, et al, 2022. PSCF and CWT methods as a tool to identify potential sources of ^7Be and ^{210}Pb aerosols in Granada, Spain. *J Environ. Radioactiv.* 251–252:106977.
- E. Chham (CA), T. El Bardouni, Y. El Mghouchi, et al, 2021. Fuel reloads optimization for TRIGA research reactor using Genetic Algorithm coupled with neutronic and thermal-hydraulic codes. *Prog. Nucl. Energy.* 133: 103637.
- E. Chham (CA), A. Milena-Pérez, F. Piñero-García, M.A. Hernández-Ceballos, J.A.G. Orza., E. Brattich, T. El Bardouni, M.A. Ferro-García, 2019. Sources of the seasonal-trend behaviour and periodicity modulation of ^7Be air concentration in the atmospheric surface layer observed in southeastern Spain. *Atmos Environ.*, 212: 148-158.
- Y. El Mghouchi, E. Chham, E.M. Zemmouri, A. El Bouardi, 2019, Assessment of different combinations of meteorological parameters for predicting daily global solar radiation using artificial neural networks. *Build Environ.* 149: 607-622.
- E. Chham (CA), F. Piñero-García, E. Brattich, T. El Bardouni, M. A. Ferro-García. 2018. ^7Be spatial pattern in the Iberian Peninsula: evaluation of a predictive model. *Chemosphere*, 205: 194-202.
- E. Chham (CA), M.A. Ferro-García, F. Pinero-García, P. Gonzalez-Rodelas. 2017. Impact of Air masses on the distribution of ^{210}Pb in the southeast of Iberian Peninsula air, *J Environ. Radioactiv.* 177:169-183.
- Y. El Mghouchi, E. Chham, M. S. Krikiz, T. Ajzoul, A. El Bouardi. 2016, On the prediction of the daily global solar radiation intensity on south-facing plane surfaces inclined at varying angles, *Energ. Convers. Manage.* 120: 397-411.
- E. Chham (CA), T. El Bardouni, K. Benaalilou, H. Boukhal, B. El Bakkari, Y. Boulaich, C. El Younoussi, B. Nacir. 2016. Neutronic and thermal-hydraulic analysis of new irradiation channels inside the Moroccan TRIGA Mark II research reactor core, *Appl. Radiat. Isot.* 116: 178-184.
- E. Chham (CA), F. Piñero García, et al, 2015. Monte Carlo analysis of the influence of germanium dead layer thickness on the HPGe gamma detector experimental efficiency measured by use of extended sources, *Appl. Radiat. Isot.* 95, 30–35. DOI: 10.1016/j.apradiso.2014.09.007.

Selected conference contributions

E. Chham, J. A. G. Orza, J. H. Christensen and Z. Ye. 2026. Long-term simulation and evaluation of cosmogenic Beryllium-7: insights into atmospheric transport and deposition processes. EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026. Poster EGU26-20989 (X5.71, 6 May). DOI: 10.5194/egusphere-egu26-20989.

- E. Chham., J.A.G. Orza., Can Weekly Measurements Efficiently Identify Areas of High Concentrations of Radioactive Aerosols?. 2024. The 8th Iberian Meeting of Aerosol Science and Technology, La Coruña (Spain) 26-28/06/2024. (Oral presentation).
- Á. C. Benito., J. A. Suárez Navarro., E. Chham., M. B. Sanz Sedano., J. F. Benavente Cuevas., A. G. Sanabria., Marta Barragán Romero., G. H. Alonso., 2021. Aplicación de una red neuronal para la caracterización radiológica de una zona impactada por minería. 46 Reunión Anual SOCIEDAD NUCLEAR ESPANOLA, Granada, 6-8 /10/ 2021. (Poster)
- E. Chham., T. El Bardouni., K. Benaalilou., B. El Bakkari., Y. Boulaich., et al, 2018. An additional irradiation channel inside the TRIGA MARK II research reactor core to improve the capacity of radioisotopes production. PHYTRA 4. Marrakech, Morocco. 17–19/09/2018. (Poster).
- E. Brattich., E. Chham., F. Piñero-García., et al, 2017. On the possible connection between African dust outbreaks and stratosphere-to-troposphere exchange as measured in the Iberian Peninsula. 5th Iberian Meeting on Aerosol Science and Technology, Barcelona, Spain. (Poster).
- E. Chham., F. Piñero-García., M. A. Ferro-García, 2016. Influence of Air Masses on ²¹⁰Pb Aerosols in the South-East of Iberian Peninsula. The 8th International Workshop on Sand/Dust storm and Associated Dust fall (Dust workshop8), Lisbon, Portugal. (Poster).
- E. Chham., F. Piñero-García., M. A. Ferro-García., 2016. Determination of the HpGe dead layer by the Monte Carlo method. The IX Conference on Quality in the control of Environmental Radioactivity. Barcelona, Spain (Oral presentation).
- E. Chham., F. Piñero-García., M. A. Ferro-García., M. Azahra., T. El Bardouni., 2015. Monitoring of ⁷Be in surface air of Granada and their variations with Solar Irradiance and meteorological parameters. The 4th Joint Congress Spanish Society of Medical Physics (SEFM) And the Spanish Society of Radiological Protection (SEPR) Valencia, Spain (Poster).

Peer review

Reviewer for Chemosphere, Heliyon and Nuclear Instruments and Methods in Physics Research Section B.

Research projects

2021/PER/00019, A new method for estimating cosmogenic Be-7 production rates and its integration into an atmospheric transport model. Generalitat Valenciana. 15/10/2024-31/10/2026. Principal Investigator.

2022/PER/00001, Study of Radionuclides and Suspended Particles in Ambient Air in Relation to the Atmospheric Dynamics of the Northern Hemisphere. 06/04/2023-31/10/2024. Principal Investigator.

IAEA Research Contract No. 26322-2022. International Atomic Energy Agency (IAEA) project: Development of neutronic and thermohydraulic coupled calculational methodologies for research reactors including treatment of uncertainties (F12028). 16.000 €. Researcher.

CGL2015-70741/CLI, Impacto de las intrusiones de masas de aire con polvo africano y de masas de aire estratosférico en la Península Ibérica. Influencia de El Atlas. Ministerio de Ciencia e Innovación. Investigación. J.A. García Orza. (University Miguel Hernández). 01/01/2016-31/12/2018. 141.000 €. Team member.

A1/041236/11, Implementation of an Environmental Radiation Monitoring Laboratory at the University of Tetouan. Agencia Española de Cooperación Internacional. M.A. Ferro García. (Universidad de Granada). 01/01/2012-31/12/2012. 21.560 €. Researcher.

Research contracts and fellowships

- Postdoctoral Researcher under the APOSTD program (10/2024 – Present)

- Distinguished Postdoctoral Researcher under the María Zambrano program (04/2023 – 10/2024) at Miguel Hernández University

Postdoctoral Researcher, ERSN Laboratory, Faculty of Sciences, Abdelmalek Essaâdi University, Tétouan, Morocco (2020–2022).

Predocctoral fellowship under the Erasmus Mundus Al-Idrisi II programme, University of Granada (10/2014–07/2017).

Teaching and supervision activities**PhD co-supervision — ERSN Laboratory, Abdelmalek Essaâdi University**

- Air-mass back-trajectory analysis tool (2020–2025): defended; two associated publications.
- Radioactive aerosols in Tétouan and Granada (2019–2025): defended; two associated publications.

Master's thesis supervision — PTR programme, Abdelmalek Essaâdi University

- PM₁₀ trends in Granada (2001–2019) (2023–2024). • ⁷Be in Granada during solar cycle 24 (2006–2017) (2019–2020).
- Transport pathways/sources of ⁷Be in Granada (2018–2019). • ⁷Be & ²¹⁰Pb in Tétouan (2017–2018).

Master's thesis committees — PTR programme

- Total α/β variation in air (2019–2020). • ⁷Be in Granada (2006–2017) (2019–2020).
- HPGe detector modelling with Compton suppression (PGAA, CENM) (2020–2021).

University teaching — laboratory sessions

- Fluid Mechanics, UMH-BSc Industrial Electronics & Automation (2023–2025: 75 h) and BSc Electrical Engineering (2023–2024: 15 h).
- Monte Carlo Simulation, Master PTR, Tétouan (2012–2014, 2018–2019: 60 h).
- Nuclear Energy, BSc Physics, Tétouan (2016–2017: 20 h).
- Radiation Detectors, Chemical Engineering, Univ. of Granada (2014–2015: 30 h).

5. Capacity of the participating organisations

5.1 Overview of the participating organisations

Role	PIC	Short name	Academic (Y/N)	Country	Supervisor
Beneficiary	999997736	AU	Y	Denmark	Dr Zhuyun Ye

5.2 Capacity of the beneficiary

Beneficiary: Aarhus University (AU), Denmark	
General description	Aarhus University is a Danish public research university. STRAT-TRACE will be hosted by its Department of Environmental Science in Roskilde. The Atmospheric Modelling group studies atmospheric transport, chemical transformation and deposition from local to hemispheric scales, linking air-pollution modelling with environmental and health assessment and decision support.
Role and profile of supervisor	Dr Zhuyun Ye, Tenure Track Assistant Professor in Atmospheric Emissions & Modelling, will be the principal supervisor, providing expertise in atmospheric chemical-transport modelling and DEHM development. Jesper Heile Christensen, Senior Researcher and developer of DEHM, will provide complementary scientific and technical mentoring. The researcher's three one-month visits in March 2025, October 2025 and May 2026 establish prior working experience with the host.
Key research facilities, infrastructure and equipment	AU-ENVS develops DEHM, which contributes to European regional air-quality production within CAMS. The project will use the host's confirmed institutional Linux/HPC resources, model-development expertise and shared storage capacity exceeding 600 TB. The controlled experiment matrix and retained diagnostics will be benchmarked, and the project allocation agreed with the host, before full production.
Previous and current involvement in EU-funded research and training programmes / actions / projects	CAMS regional air-quality production with DEHM (ongoing); CAMEO, CAMS Evolution (2023–2025, completed); MARCHES (2023–2026, coordinated by AU), on the health costs of environmental stressors; GreenEO (2025–2029, AU partner), on Earth-observation-enabled environmental modelling and services.

6. Additional ethics information

STRAT-TRACE uses atmospheric modelling and existing environmental observations; it does not involve human participants, personal health data, animal experiments or new releases of radioactive material. Reuse of third-party observations will follow the relevant access agreements, licences and data-management procedures. CTBTO data are subject to an agreement that has been signed, with delivery still pending; access restrictions and permitted redistribution will be respected. Open outputs will comprise code, methods and data products that can lawfully be shared.

7. Additional information on security screening

The project pursues civilian environmental research on atmospheric transport and aerosol removal. No classified information is anticipated. Restricted access to third-party observations, including the CTBTO dataset planned under the signed agreement, will be managed through authorised access, appropriate institutional storage and the applicable contractual conditions. Restricted access is not treated as evidence that these observations are classified; any specific security obligations will be checked with the host before data use.

8. Environmental considerations in light of the MSCA Green Charter

STRAT-TRACE will reuse existing atmospheric observations and modelling resources, with no new field campaign or radioactive releases. Its environmental footprint will be managed through proportionate computing, data stewardship and travel choices.

The DEHM experiment matrix will be benchmarked before full production. Shared inputs, restart files and validated intermediate products will be reused, and unnecessary repeat runs avoided. Storage will prioritise documented daily and monthly diagnostics and selected three-dimensional fields needed for reproducibility, within the agreed institutional computing and data-management arrangements.

Routine collaboration and suitable dissemination activities will use online meetings. Essential research visits and conference travel will be planned together where feasible, with rail preferred when practical. Documentation will be digital first; code and reusable outputs will be shared in interoperable formats where licences permit.